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SYMPOSIUM PAPER

Deepak P. Assudani · Roger B. V. Horton
Morgan G. Mathieu · Stephanie E. B. McArdle
Robert C. Rees

The role of CD4+ T cell help in cancer immunity and the formulation of novel cancer vaccines

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Abstract Recent years have seen the unprecedented surge of interest in the role of CD4+ T cells and the role they play in the development of the immune response. In this symposium review, we examine the evidence for this and discuss their functions, particularly in respect to the cancer immunology, including CD4+CD25+ cells (Treg).

Are CD4+ T cells really important?

Earlier studies established the undisputable role of CD8+ T cells in the recognition and eradication of tumour cells. Most of these studies used adoptive transfer of tumour-specific CD8+ T cells in tumour-bearing mice and demonstrated regression. Immunisation of these animals with MHC class I peptides has also been shown to induce tumour regression. These models undermined the role of the CD4+ T cells. Indeed, when these therapies were transferred to the clinic; despite the generation of cytotoxic T lymphocytes (CTL), clinical response was rarely observed [21, 95]. In other animal models it was observed that the CD4+ T cells play a central role in orchestrating the immune response against the tumour cells as well as having a direct role in tumour rejection. In our studies, using a disabled single cycle–herpes simplex virus (DISC–HSV) in a CT26 tumour regression model, we showed that treating mice with established CT26 tumours using DISC–HSV virus led to regression of tumours in about 70% of the animals and that both CD4+ and CD8+ T cells were important for efficient tumour regression [3, 5, 6, 61, 66]. Realising the importance of CD4+ T cells, efforts to identify MHC class II restricted peptides have intensified, leading to discovery of several naturally processed class II peptides by us and others, which could potentially be ideal partners to MHC class I peptides in a peptide vaccine cocktail [68, 90, 104].

In various tumour models, CD4+ T cells have been shown capable of mediating tumour regression on their own [23, 27, 50]. Models of CD4+ T cell knockout mice and adoptive transfer of CD4+ T cells have clearly demonstrated the ability of CD4+ T cells to mediate tumour rejection without CD8+ T cells [36, 78]. Studies in severe combined immunodeficient mice showed that human CD4+ T cells generated from PBMC of lung cancer patient could mediate rejection of autologous lung tumour xenografts in them despite tumour cells being HLA negative [30]. Tumour rejection by CD4+ T cells was mostly mediated directly by IFN-γ-dependent mechanisms, whereas in other models it has been shown to be through the recruitment of macrophages and/or eosinophils [94]. IFN-γ has indeed been shown to have pro-apoptotic and anti-proliferative effect on tumour cells and was also shown to inhibit angiogenesis by inducing up-regulation of MHC class I molecules, making tumour cells more susceptible to CTL mediated killing [42]. Most studies suggest that CD4+ T cell-mediated tumour rejection is not MHC restricted [30]. However, some evidence suggests that CD4+ T cells can recognise tumour cells and lyse them in an MHC restricted fashion [87]. Independent studies have shown CD4+ T cells to directly mediate tumour cell lysis through TRAIL, FasL and granzyme–perforin dependent pathways, traditionally employed by CTL [29, 74, 87, 88]. However, this is unlikely to be the primary role for CD4+ T cells as most tumours are MHC class II negative [45, 52]. Thus, although CD4+ T cells can mediate tumour rejection directly on their own, their main role involves orchestrating the immune response

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D. P. Assudani · R. B. V. Horton · M. G. Mathieu
S. E. B. McArdle · R. C. Rees (✉)
School of Biomedical and Natural Sciences,
Nottingham Trent University, Clifton lane,
NG11 8NS Nottingham, UK
E-mail: robert.rees@ntu.ac.uk

through CTL, macrophages and eosinophils in certain cases.

Clinical trials using adoptive transfer of CTL in cancer patients have also reinforced the value of CD4+ T cells. Clinical trials using CD8+ T cell clones have shown that the transferred T cells could persist for prolonged periods of time provided that the antigen burden is low, however these cells rapidly disappeared if higher levels of antigen were present [91, 93, 103]. Co-transfer of CD4+ T cell help with the CD8+ T cells has been shown to prolong the survival of adoptively transferred TIL [13, 103]. Therefore CD4+ T cell help is required for in vivo survival as well as expansion of tumour-reactive CTL ex vivo. Some early work showed that it was possible to isolate tumour antigen-specific CD4+ T cells from tumour infiltrating lymphocytes in melanoma patients [89]. Adoptive transfer into patients of unfractionated T cells (CD4+ and CD8+ T cells) has been shown to promote tumour regression [70]. Such early studies indicated that the full activation of auto-reactive CD4+ T cells was likely to be an important component that is currently missing from many clinical trials.

Dudley and Rosenberg have pioneered work in the field of adoptive T cell transfer in the clinic. Results from the most recent trials support the notion that CD4+ T cells are critical for efficient immunotherapy. In early trials, cloned melanoma-reactive T cells were adoptively transferred into 13 patients with metastatic melanoma followed by administration of IL-2 [27]. These transferred cells had high reactivity against a number of melanoma antigens when tested in vitro and yet not a single objective response was seen in any of the 13 patients. In the second trial, patients with metastatic melanoma were given cloned high avidity T cells that were transferred post-non-myeloablative chemotherapy [28]. This study utilised increasing doses of cyclophosphamide and fludarabine in conjunction with IL-2. No objective clinical responses were seen in 15 patients tested [28].

In both trials it was speculated that the poor persistence of the adoptively transferred clones was responsible for the lack of clinical responses. Indeed just 2 weeks post-re-infusion none of the transferred cells could be detected in the peripheral blood of treated subjects. Patients received CD8+ cloned T cells, which underwent multiple in vitro restimulations in the presence of OKT3 and IL-2. It was hypothesised that these non-physiological culture conditions in conjunction with a lack of CD4+ T cell help lead to the failure of the transferred cells to become activated and persist in vivo. As a result the protocol for a subsequent clinical trial was modified to administer a heterogeneous population of TIL containing both CD4+ and CD8+ T cells [69]. In this ongoing trial 35 patients have been treated, 18 of which (51%) have generated an objective clinical response, including four patients with a complete response. Thus administration of lympho-depleting chemotherapy in conjunction with a minimally cultured heterogeneous T cell population may be responsible for this increase in objective response rate. Moreover T cells were shown to persist for extended periods of time (up to 2 years) and this persistence correlated with clinical responses [69].

Although CD4+ T cells will undoubtedly prove useful in the clinic, perhaps the question should be asked as to the precise nature of T cell help required in order to further improve clinical efficacy of tumour-specific CTL. Is non-specific help, for example virally derived antigen such as Hepatitis B core antigen, the most effective way to promote CTL response or would protein-specific T cell help, i.e. the CTL epitope and the helper epitope derived from the same protein prove more effective? Preliminary transgenic mouse (C57bl/6-HHD II) studies performed in our laboratory utilising human class I p53 peptides in combination with either HepB or mouse class II p53 peptides indicated that the latter is more effective and can boost both number and magnitude of CTL responses to a particular antigen in vitro (R. Horton, unpublished data). Theoretically, if a pre-existing response to a class I restricted tumour antigen is present then there may also be a class II response to the same antigen. By immunising with class I and class II peptides derived from the same protein against which there is a prior immune response, it would seem logical that a stronger anti-tumour response could be generated. The theory is that an anti-tumour response could be invoked in the same manner as a secondary immune response to a pathogen that has previously been encountered. Further study is necessary in order to confirm these early novel in vitro findings as well as investigating the effects of T-helper peptides in an in vivo tumour model. Moreover, nature of the T cell help might be dependant on the peptides which might dictate generation of either Th1 or Th2 response. It is widely believed that cytokines secreted by Th1 cells facilitate CTL generation whereas Th2 cytokine profile might favour antibody response and be detrimental for CTL induction [47, 58].

CD4+ T cells: to help or not to help and when?

Although most of the studies published to date suggest that CD4+ T cell help is required for the generation of memory CTL responses, some studies indicate that they are also essential in the priming phase of the CD8+ T cells [12, 17, 33]. However, other studies have shown that naïve CD8+ T cell priming does not require CD4+ T cell help, especially in case of pathogenic infections [15, 79, 84, 99]. It has been argued that infection by pathogens provide strong “danger signals” to the APC by stimulating their “Toll like Receptors”. This leads to up-regulation of the CD40 molecules without the need for the CD4+ T cell priming, thus bypassing the need for the helper requirement in certain cases. However, this hypothesis does not explain models where no pathogenic infections are involved. Recent studies in tumour models have demonstrated CD4+ independent generation of CTL and effective clearance of established tumours, even though no inflammatory signals were necessarily generated [62, 105]. Thus depending on the model used, CD4+ T cells may or may not be required for priming of initial CTL reaction. Also, different immunisation protocols might have an important role to play in the generation of CTL.

CD4+ T cell help might be dependent on the MHC class I epitope affinity. Franco et al. [31] showed that peptides with high affinity for their MHC class I molecules do not require CD4+ T cell help. They observed that immunising mice with three different high affinity MHC class I epitopes generated CTL, irrespective of T cell help, whereas immunisation with subdominant epitopes required CD4+ T cell help for generation of CTL. The nature of help required to generate cytotoxic response itself was dependent on the class I peptide [31]. In the same way, Rosenberg et al. [71] showed that melanoma patients immunised with a modified peptide, with increased binding affinity for HLA-A2.1 molecules, was able to generate CTL in most patients and resulted in clinical responses in 42% of them. No helper peptide was used in this study. These studies suggest that the generation of CTL is influenced by the period of MHC restricted peptide display on APC, TCR–MHC binding affinity/duration and whether or not CD4+ T cell-mediated help is necessary. The formation of long-lasting MHC–peptide–TCR complexes depends upon the binding affinity of the peptide and establishes a cross-talk between CD8+ T cell and the APC. This ultimately leads to activation of the APC and CTL generation in the absence of CD4+ T cell help. In support of this, it has recently been shown that CD8+ T cells can condition the APC in an IFN-γ-dependent mechanism [57, 72]. Also, the presence of high antigen concentrations can lead to CD8+ T cell activation without CD4+ T cells [48]. However, any conclusions drawn from transgenic mouse models or adoptive therapy experiments have to be considered with caution as such data may not be directly applicable in humans.

Although naïve CTL priming in some circumstances does not require CD4+ T cell help, it has been well established that these cells are vital for the generation of memory CTL [79]. CD4+ T cells seem to be important in the maintenance of CD8+ T cell memory and facilitate their survival, although some studies suggest that memory CTL response only requires CD4+ T cell help at the initial priming moment [75, 79, 96]. In specific circumstances, a primary CD8+ T cell response might be generated effectively in the absence of a helper response. In contrast, the memory response to secondary challenge upon re-exposure of animals to antigen is defective suggesting a critical requirement for CD4+ T cells in the initial memory-priming phase [79, 84]. However, most studies have been conducted utilising pathogens and whether the mechanism in cancer patients is similar is not entirely clear. Recently, Gao et al. [33], using a transgenic mouse model and ovalbumin as a tumour antigen, showed that helper T cells are required for re-activation of memory T cells into functional tumour killer cells. Other studies contradict these findings and further work is required to determine precise mechanisms [12]. The requirement for CD4+ T cell help in the generation of optimal CTL responses might be dependent upon the antigen and the MHC class I epitope. However, because of their ability to enhance the anti-tumour CD8+ T cell response, many groups, including ourselves, believe that the generation of tumour-specific CD4+ T cells should be an essential component of any clinical trial.

Mechanisms of T cell help

One of the intriguing questions has been the mechanism of the help provided by the CD4+ T cells to the APC and/or CD8+ T cells. Several lines of evidence argue that different mechanisms exist for CD4+ T cell help, consisting of CD40-dependent and CD40-independent pathways. CD40-independent help was initially thought to be due to release of IL-2 from activated CD4+ T cell which would then activate CD8+ T cells in the vicinity [49]. Other studies showed that there might be another mechanism responsible, where APC are primed by CD4+ T cells, which would then prime CD8+ T cells. This mechanism was shown to be dependent on the CD40L expression on activated CD4+ T cells, which binds to the CD40 receptors on the surface of APC and licenses them for effective CD8+ T cell activation. This model was supported by experiments showing that CD4+ T cell help can be bypassed by direct stimulation of the CD40 receptors on the dendritic cells (DC), thereby replacing the signal provided by CD4+ T cells [11, 76].

Studies have also shown that help by CD4+ T cells can be mediated in a CD40-independent manner. Using influenza hemagglutinin as an antigen model, Lu et al. [49] showed that priming of CD8+ CTL required CD4+ T cell help but this help could be generated in CD40 −/− mice or could not be bypassed by using CD40 antibodies. Moreover, help was not found to be mediated by lymphokines, which suggested that it may have been a cell–cell interaction mediated mechanism. It has been hypothesised that the TNF-related activation-induced cytokine (TRANCE) receptor might be the candidate cell surface molecule involved in providing CD40 independent help by CD4+ T cells [9, 49]. CD8+ T cells have been shown to be able to mature DC on their own in a CD40 independent manner [72]. Intriguingly, CD8+ T cells have been shown to express CD40 molecules transiently after activation and this is necessary for them to receive CD4+ T cell help for memory CTL generation [14]. NK cells also possess the ability to activate DC and generate CTL in a CD4+-independent manner [1].

Recently, it was shown that CD4+ T cells control the generation of memory CD8+ T cells via a TRAIL-dependent mechanism. CD8+ T cells primed in the absence of CD4+ T cell help, up-regulated TRAIL and upon secondary encounter with the antigen underwent activation induced cell death (AICD), whereas those primed in presence of help did not up-regulate TRAIL and were able to undergo significant clonal expansion on re-encounter of antigen [43]. This potentially represents a complex mechanism through which the immune system avoids autoimmunity, as most of the CD8+ T cells generated in vivo to self-antigens would occur in the absence of help. Thus on re-encountering the same antigen, T cells would undergo AICD. This could also explain the limited success in generating high affinity cancer antigen-specific CTL, as high affinity CTL would undergo AICD.

Several studies have established that for optimum CTL generation, CD4+ and CD8+ T cells have to recognise the antigen on the same APC [67, 80]. Using a chimeric mouse model consisting of two different types of DC, one which could only activate CD4+ T cells and other which could only present antigen to CD8+ T cells, Smith et al. [80] showed that immunising the mice generated poor CTL response. Therefore, an optimised CTL response is achieved after cognate recognition of both class I and class II epitopes respectively by CTL and T-helper cells on the same DC. However, an intriguing question for immunologists has been whether it is necessary for the CD4+, CD8+ and the APC to come into contact with each other at the same time or not. Several studies have provided evidence both for and against the requirement for simultaneous stimulation of CD4+ and CD8+ T cell by the APC.

It must be remembered that the frequency of naïve antigen-specific precursor T cells is normally very low, hence, antigen carrying APC encountering such rare antigen-specific CD4+ T cells and equally rare CD8+ T cell at the same time seems highly unlikely, although it has been argued that contact between APC and CD4+ T cells could be long-lasting, providing enough time for the antigen-specific CD8+ T cells to encounter this complex. Ridge et al. [67] showed that DC can first engage CD4+ T cell and the cross-talk between them through CD40–CD40L interaction licenses the DC to then activate naïve antigen-specific CD8+ T cells at a later stage. This two step model seems more realistic as the rare antigen-specific CD8+ and CD4+ T cells do not have to come into contact with few DC carrying the specific antigen at the same time. Recently, another two-cell dynamic model has been suggested by Xiang et al. [100]. Their study reveals that during the initial encounter of antigen carrying APC and CD4+ T cells, along with the MHC-class II peptide complex and co-stimulatory molecules, the CD4+ T cells also acquire the MHC-class I peptide complexes from the DC. These activated CD4+ T cells can then themselves act as APC for the naïve CD8+ T cells by virtue of displaying MHC class I peptide complex along with co-stimulatory molecules. Moreover, these CD4+ APC were able to effectively generate CTL and anti tumour immune response in a murine tumour model [100]. It has also been suggested that the various pathways of CD4+ T cell help are dependent on the epitopes [31]. It seems possible that different help mechanisms act synergistically for optimum immune activation. However, it might be interesting to determine the threshold of MHC–peptide–TCR interaction required to generate different help patterns and whether CD40-dependent and CD40-independent pathways are a function of such threshold of T cell receptor (TCR) activation. Considering the above, it would seem that any cancer vaccination protocol which is unable to generate optimum CD4+ T cell help is unlikely to succeed as most of the T cells for high affinity epitopes are likely to have been deleted/tolerised by either central or peripheral tolerance mechanisms [37]. Many of the remaining T cells would be of medium to low binding affinity, which would require help for optimum activation and memory responses.

Seeing is believing

Up until a few years ago it was not possible to visualise the immune system in vivo. However within the past few years, a technology known as intra-vital imaging was developed and allows tissues, organs and even single cells hundreds of micrometers below the surface of an anaesthetised animal to be visualised in real time. This has been particularly useful in the study of the immune system where T cells and dendritic cells can be tracked in the lymph node in order to monitor cellular behaviour [34, 39]. In this way the interaction between CD4+ T cells and DC and its effects on CD8+ T cells can be studied visually for the first time. Antigen-specific T cells are extremely rare and these must come into direct contact with the marginally more abundant antigen loaded DC in lymph nodes (LN) draining from the periphery. The dendrites on DC increase the volume of LN that a DC can sample by around threefold [53]. Consequently, T cells coming into contact with the tip of a DC dendrite change morphology and move into close contact with the DC to initiate priming.

In order for T cells to come into contact with DC, a homing mechanism must be in place. Innate receptor signals provided by TLR binding in conjunction with antigen encounter induce DC to secrete chemokines that in turn are recognised by receptors on memory T cells [86]. Such activation of memory T cells in conjunction with CD4+ T cells is critical in cancer patients to prevent disease relapse [32]. This activation induced chemokine secretion implies that a cytokine gradient mediated T cell migration may occur in the LN. Germain et al. [34] immunised mice with ovalbumin loaded DC and subsequently injected T cells specific for ovalbumin peptides (CD8+ OT-I and CD4+ OT-II). In the presence of OT-II T cells, OT-I and polyclonal CD8+ T cells aggregated around OT-II specific peptide pulsed DC and showed a 3–5-fold increase in contact frequency compared to interactions with unpulsed DC (both $P<0.001$). Further investigation of the underlying mechanism revealed that prior to antigen recognition, naïve CD8+ T cells up-regulated CCR5 expression when in an inflammatory environment. This allowed them to be efficiently attracted to sites of antigen-specific interaction between CD4+ T cells and DC under the guidance of the chemokines CCL3 and CCL4. Antibody blocking experiments demonstrated the importance of the CD4+ T cell/DC interaction in the generation of CD8+ T cell memory as interference with this interaction severely reduced the ability to form functional memory CD8+ T cells in a vaccine model [38].

CD4+ T cell help is therefore crucial for the activation of DC, which then produce chemoattractants critical in facilitating interaction between CD8+ T cells and DC that induce clonal expansion of rare antigen-specific T cells. This is even more important in the tumour environment as the influence of CD4+ T cells could be responsible for the breaking of tolerance leading to tumour clearance [82].

These studies have to be viewed with caution as they have been conducted in adoptive transfer models or transgenic TCR models with high frequency of antigen-specific T cells.

Tolerance, T-regulatory cells and their control

It is generally believed that tolerance is a consequence of anergy, exhaustion, ignorance, peripheral deletion or immunosuppression. Lymphocytes can remain alive for a long period of time but functionally hyporesponsive after an antigen encounter. This anergy can either be clonal under a growth arrest state after an incomplete activation or adaptive as proliferation and effector functions are inhibited following a lack of co-stimulation or an excess of co-inhibition [77]. Foreign antigens such as viruses may persist after a primary infection. Indeed, CD8+ T cells may be induced by an acute viral infection, expand all at once and die from exhaustion within a few days allowing the remaining viruses to develop and be tolerated by the immune system [54]. Ohashi et al. [59] generated transgenic mice expressing a lymphocytic choriomeningitis viral (LCMV) glycoprotein in the pancreas and compared them with non-transformed mice. The former group reacted more readily to LCMV infection than the latter proving that ignorance may lead to the expansion of an unresponsive self-reactive T cell population. Tumour cells can also evade immune responses using peripheral deletion by expressing at their surface CD95L (also known as Fas ligand) and triggering the death by apoptosis of T cells through a Fas/Fas ligand interaction [83]. Finally, cancer cells can produce a broad variety of immunosuppressive cytokines such as IL-6, which is released by most melanoma cells in advanced stage lesions [48]. The involvement of T-regulatory cells (Treg) as part of peripheral tolerance mechanisms complicated the picture and remains controversial [2]. Here we specifically review the role of Treg, the major subset of regulatory T cells, in tolerance against tumour and tumour immunotherapy as well as recent developments on the control of Treg made to help the action of cytotoxic and helper T cells against tumour cells.

Recent progresses in deciphering mechanisms of action of Treg

Treg population accounts for 2–5% of the total population of CD4+ T cells in both rodents and humans [19, 101]. Treg cells are generally characterised by a CD4+CD25+ phenotype. They can either be natural following selection in the thymus as an anti-self-repertoire or induced in the periphery with expression of the CD25 marker upon the action of immunosuppressive cytokines such as IL-10 or TGFβ or both [92]. Contrary to mice, CD25 expression in humans is variable [10, 35, 51, 101] with a report describing a correlation between levels of CD25 expression by peripherally induced Treg and the severity of the histological grade of a tumour [85]. On the other hand, another Treg-specific marker is nowadays used to identify Treg and it relies on the expression of the transcription factor foxp3. Its expression is undetectable in other lymphoid or non-lymphoid cells and is strongly linked with the immunosuppressive activity of natural or adaptive Treg [20]. A third and controversial population of foxp3-negative IL-10-secreting Treg has also been identified. They have been observed following high antigenic exposure but the exact role of this subset remains unclear [101]. Foxp3 on its own is enough to induce an immunosuppressive response in mice. On the otherhand, mechanisms in humans are more complicated and even the simultaneous injection of the two isoforms of foxp3 is not enough to completely down-regulate the immune response suggesting that other factors such as target antigens are required for the generation of Treg [7].

Antigen specificities and molecular mechanisms of these Treg are unknown. Wang and colleagues [97] managed to isolate LAGE1-specific Treg in a tumour-infiltrated lymphocyte line generated from tumour samples of a cancer patient. A new gene encoding for a tumour-specific self-peptide called antigen recognised by Treg cells or ARTC1 was also identified [98]. ARTC1-peptide activated Treg can suppress both proliferation of melanoma-reactive T cells and production of IL-2 meaning that local and tumour-specific suppression may be due to Treg activated by a whole new range of peptide ligands specifically presented by tumour cells. These results suggest the idea that these Treg may come from CD4+CD25− T cells which have been converted to CD4+CD25+ Treg because of the surrounding suppressive cytokines present at the tumour sites or after direct interaction with naturally occurring Treg according to the "infectious" tolerance theory [25, 65]. This is subject to controversy as another subset of

CD3+CD4+CD25− T cells identified by our laboratory present in the non-adherent fraction of parenchymal splenocytes has been shown to be strongly linked with progression of tumour in an established colon carcinoma CT26 tumour model in Balb/c mice immunised with DISC–HSV engineered to secrete mGM–CSF [2]. Indeed, increase in the survival of treated mice was achieved by delaying tumour growth and inducing complete tumour regression in 70% of mice treated. On the other hand, in progressor mice, lymphocytes from the parenchymal fraction of the spleen were capable of negatively regulate AH-1 peptide-specific CTL against CT26 tumour cells. And contrary to the infectious tolerance theory developed by Dieckmann and colleagues relying mainly on the secretion of IL-10 by anergised CD4+CD25− T cells to restrain syngenic CD4+ T cells [10, 25], the inhibition of the CTL activity by these suppressive CD4+CD25− T cells occurred in a cell-to-cell contact fashion (M. Ahmad, unpublished results) like CD4+CD25+ Treg [92]. Indeed, suppressive activity of Treg is generally admitted to occur by cell-to-cell contact. Following observations of Treg in tumour models and in humans, they can deprive auto-reactive T cells of autocrine IL-2, trigger the activation of the indoleamine 2,3-dioxygenase in DC via an inhibitory molecule called cytotoxic T lymphocyte-associated antigen 4 (CTLA-4) causing immunosuppression of T cells, down-regulate the expression of co-stimulatory molecules such as CD80 or CD86, or secrete the immunosuppresive cytokine TGFβ following engagement of CTLA-4 [51]. These results add complexity to the panel of Treg but evidences of their involvement in the anti-tumour response accumulate and imply a careful study of their mechanisms of suppression for a better understanding.

Evidence of T-regulatory cells involvement in anti-tumour response

Indeed, Treg have been extensively studied in cancer patients. Treg are mainly present in lymphoid tissues. There are also increasing evidence of their presence within peripheral blood and tumour-infiltrated lymphocytes [41]. Curiel et al. [22] demonstrated the link between presence of Treg in individuals with ovarian cancer and the decrease of tumour-specific CTL activity as well as their contribution to tumour growth. CCL22 chemokine is constantly produced by tumour cells and surrounding macrophages mediating trafficking towards the tumour making the latter appear like an immuno-privileged organ to the immune system. Moreover, it was considered that Treg could not be found in draining LN in later cancer stages [40]. Kawaida et al. [44] then showed the correlation between the infiltrations of CD4+CD25^{high} in LN surrounding the gastric tumour with higher rate of infiltration than in control distant LN and impaired cell-mediated immunity in cancer patients.

Treg have been also pointed out as the major reason for failed immunotherapies. Using a CT26 colon carcinoma model in Balb/c mice, depletion of Treg allowed the induction of both CD4+ and CD8+ anti-tumoral response. By depleting both Treg and CD8+ T cells, CD4+ T cells were also capable of rejecting MHC class II-negative CT26 tumour cells by causing IFN-γ-dependent anti-angiogenic activity. Moreover, restoration over time of the Treg population immediately repressed this anti-tumour response [16]. Vaccination relying on GM–CSF gene transferred to autologous tumour cells was proven to be efficient in some subjects with severe cases of cancer such as advanced melanoma, non-small cell lung carcinoma, ovarian carcinoma or myeloid leukaemia. T- and B-lymphocytes were found to be densely infiltrated in distant metastases achieving extensive tumour necrosis [73, 81]. However, despite obvious signs of improved immunity, the majority of vaccinated patients eventually evolve to progressive disease [26]. More recently, a study showed that a CTL response was obtained after immunisation of melanoma patients with MHC class I peptide- or melanoma tumour lysate-pulsed antigen presenting cells which peaked at 7 days post-immunisation. The CTL response was then in decline reaching the pre-vaccine level 28 days after administration of the vaccine. This decrease correlated with the expansion of the IL-10-secreting Treg population in post-vaccine peripheral blood lymphocytes [18]. These results in mice and humans emphasise a similar involvement of Treg in both species, which is the maintenance of tolerance and the down-regulation of the anti-tumour response.

The critical issue is to find a balance between maintenance of tissue integrity by avoiding autoimmune response to take place and induction of a potent tumour-specific immune response allowing successful tumour regression. The first point must be of major consideration as immunotherapeutic targets are often normally expressed in most healthy tissues and abnormally or over-expressed in cancer tissues. In fact, a large part of the work carried out in our laboratory relies on the study and the eventual therapeutic utilisation of tumour antigens derived from a tolerated non-dangerous antigen repertoire such as the tumour protein p53 with the discovery of MHC class II p53-derived peptides using HLA-DR1- and HLA-DR4-transgenic mice, or the metastasis-associated antigen MTA-1, an antigen widely over-expressed in murine and human tumours currently under investigations for its immunogenecity and its potential as a tumour rejection antigen [8, 68]. It is therefore necessary to discover new techniques or agents capable of overcoming or reversing T cell tolerance to tumour antigens.

Proposed methods to counteract Treg activity

There have been few proposed methods to counteract Treg and Fig. 1 summarises some of them. Blockade of CTLA-4 appeared to be a promising idea. It is an important immunoregulatory molecule expressed at the surface of activated T cells and Treg. It interferes with TCR signalling and lead to the down-regulation of T cells. Like CD28, CTLA-4 competes with the latter to bind B7 molecules present on the surface of antigen-presenting cells but with a higher affinity. It also induces specific tolerance to allo- and self-antigens by interacting with the co-receptor of B7 molecules [60]. CTLA-4 antibody (MDX-010) was administered to melanoma patients concomitantly with an MHC class I restricted peptide from gp100 melanoma-associated antigen. Out of 14, three patients showed objective cancer regression but six patients also developed grade III/IV autoimmune reactions [64]. Glucocorticoid-induced tumour necrosis factor receptor family-related protein (GITR) is another constituvely highly expressed marker on Treg surface as indicated by evidence that agonistic anti-GITR antibody blocks or reduces the suppressive activity of Treg [101]. Ko and colleagues [46] showed that anti-GITR antibody treatment leads to regression in tumour-bearing mice with a large population of CD4+ and CD8+ T cells, including IFN-γ-secreting cells infiltrating regressing tumours. Activated T cells also express strongly GITR but their activity does not seem to be affected by the treatment. The latter might eventually make the effector cells capable of overcoming the suppressive activity of Treg. Interestingly, synergistic effects were also observed when both agonistic anti-CTLA-4 and anti-GITR were co-administrated in tumour-bearing mice allowing rejection of more advanced tumours. Therefore, CTLA-4 and GITR blockade could be use for cancer immunotherapy but further investigations are required to prevent their side effects.

Another novel approach is to target the IL-2 receptor CD25 subunit using the recombinant IL-2 diphteria toxin conjugate DAB_{389}IL-2 also called ONTAK. This drug specifically kills and depletes CD25-expressing Treg from the PBMC of cancer patients through the diphteria moiety without any noticeable toxicity to other populations of cells. In one study, patients with renal cell carcinoma were first treated with a dose of ONTAK in the pre-vaccination period to avoid inhibition of DC-mediated activation of T cells and then immunised 4 days later with a RNA-transfected DC vaccine. Population of Treg was dramatically reduced and correlated with a major increase in tumour-specific cytotoxic T cells [24]. Moreover, Ahmadzadeh and Rosenberg [4] demonstrated that high dose of IL-2, known as a major T

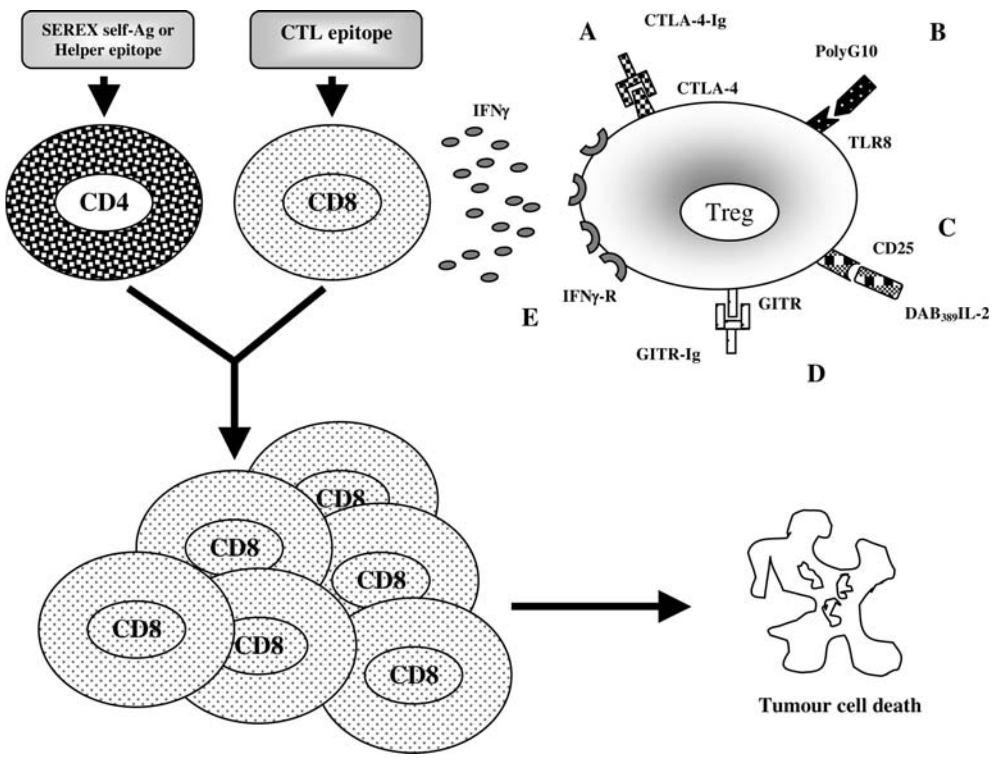

Fig. 1 Mechanisms of inhibition of Treg suppressive activity and enhancement of cytotoxic and helper T cell activity. **a** Blockade of CTLA-4 using anti-CTLA-Ig. **b** Reversal of TLR8 using CpG-A or polyG10. **c** Elimination of Treg using the recombinant IL-2 diphteria toxin conjugate DAB_{389}IL-2. **d** Blockade of GITR using anti-GITR-Ig. **e** Production of IFN-γ by cytotoxic T cells inhibiting Treg activity following co-immunisation with SEREX-defined self-antigen and CTL epitope. Any of these five mechanisms allows blocking of Treg immunosuppressive activity while immunisation with a CTL epitope accompanied with a helper peptide triggers clonal expansion of antigenic peptide-specific cytotoxic T cells leading to tumour cell death

cell growth factor, considerably enhanced the number of circulating Treg in patients. Therefore, selective inhibition of IL-2 receptors expressed on the surface of Treg using ONTAK may help in the improvement of IL-2-based therapies.

Contrary to peptide vaccination or loaded antigen-presenting cells, viral vaccination constantly provides TLR signals and allows the bypass of regulatory T cell-mediated tolerance [102]. Therefore, by finding a ligand capable of interacting with one of these TLR involved in the regulation of the T cell function, Treg function could be overcome or reversed. These innate receptors react to microbial products such as LPS, polyI:C or CpG, allowing the maturation of DC and production of cytokines triggering both innate and adaptive immune response. Peng and colleagues [63] have therefore compared different TLR ligands and observed proliferation of naive CD4+ T cells in the presence or absence of DC and Treg. CpG-A and a new ligand made of a stretch of guanosine-containing oligonucleotides called polyG10 were identified as best ligands for an almost complete reversal of the TLR8–MyD88–IRAK4 signalling pathway in Treg and the restoration of the proliferation. Moreover, the recent finding that different TLR ligands can work synergistically as adjuvants for Th1 responses [55], indicates that the administration of two agonists of TLR may help in both reversal of Treg function and enhancement of T cell response.

Interestingly, Nishikawa et al. [56] revealed by serological recombinant expression (SEREX) a new set of wild type non-mutated antigens that are specifically recognised by CD4+CD25+ T cells as well as T-helper cells. Enhancement of pulmonary metastasis after challenge with syngenic tumour cells and acceleration of tumour development of 3-methylcholantrene-induced primary tumours were observed after immunisation with DNA encoding for these SEREX-defined self-antigens. These effects were specifically due to the involvement of Treg suppressing the activity of invariant NKT and NK cells and the proliferation of CD4+CD25- and CD8+ T cells. On the other hand, when co-immunisation was performed with DNA encoding for the complete SEREX-defined self-antigen and a tumour-specific CD8+ T cell epitope, resistance to tumour challenge was heightened in a CD4+ T cell dependent manner with increased cytotoxic responses. They later demonstrated that this mechanism occurred through production of IFN-γ by CD8+ T cells abrogating generation/activation of CD4+CD25+ T cells and allowing both cytotoxic and T-helper cells to act normally against tumour development [57]. This study offers a new prospect in term of tumour immunotherapy by proposing a new way of designing cancer vaccines. It also emphasises the fact that the utilisation of over-expressed self-antigens, although attractive for tumour immunotherapy, can trigger the activation of Treg. Care must therefore be taken in the construction of a cancer vaccine by eventually incorporating IFN-γ to prevent this unwanted phenomenon or by finding an MHC class II epitope derived from these SEREX-defined self-antigens and specific for CD4+ T-helper cells only.

In view of the results of the recent clinical trials, a successful control of the Treg population appears to be the link with inefficient cancer vaccine and potent long-lasting anti-tumour response ending up eventually with tumour eradication. Therefore, more work need to be done to identify clearly the different subsets of regulatory T cells, understand their interactions with effector cells, clarify the molecular mechanisms behind the maintenance of tolerance to tumour cells and boost the action of cytotoxic and helper T cells against tumour cells.

Conclusion

The requirement for CD4+ T cell help in the field of cancer immunotherapy is well accepted. Not only can they lead to tumour eradication on their own but their main importance lies in their ability to activate other cells along with generation of a long-lasting memory CTL response. We and others have been working for the identification of HLA-DR restricted peptides which would hopefully be incorporated into future cancer vaccines. However, caution will have to be exercised when choosing the appropriate class II antigen as injecting CD4+ T cell epitopes from certain antigens or in the wrong context could lead to generation of Treg, proving detrimental. Eventually an effective cancer vaccine, containing antigens to activate CTL, will have to activate CD4+ T cells and inhibit the CD4+CD25+ T cells at the same time.

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